DSBC-AM ../6

$$v_{am}(t) = V_c \cos \omega_c t + \frac{mV_c}{2} \cos(\omega_c + \omega_m)t + \frac{mV_c}{2} \cos(\omega_c - \omega_m)t$$

- Components of $v_{am}(t)$ in phase,

$$V_{\max} = V_c + V_{USB} + V_{LSB}$$

DSBC-AM ../7

- If 2 or more sine or cosine waves modulate a single carrier
  - ❖E.g.

$$v_c(t) = V_c \cos \omega_c t \quad and \quad v_m(t) = V_{m_1} \cos \omega_{m_1} t + V_{m_2} \cos \omega_{m_2} t$$

$$v_{am}(t) = V_c \cos \omega_c t + \frac{V_{m_1}}{2}\left[\cos(\omega_c + \omega_{m_1})t + \cos(\omega_c - \omega_{m_1})t\right] + \frac{V_{m_2}}{2}\left[\cos(\omega_c + \omega_{m_2})t + \cos(\omega_c - \omega_{m_2})t\right]$$

DSBC-AM ../8

- Modulation index

$$m_1 = \frac{V_{m1}}{V_c}, \qquad m_2 = \frac{V_{m2}}{V_c}, \qquad m_3 = \frac{V_{m3}}{V_c}$$

- Total modulation index

$$m_T = \sqrt{m_1^2 + m_2^2 + m_3^2 + \ldots}$$

Exercise 1x

- A 12V, 2 MHz carrier is amplitude modulated by the signal

$$v_m(t) = 4\sin 2000\pi t + 2\sin 4000\pi t + 1.5\sin 6000\pi t$$

Determine

(a) the amplitude and frequency of each component in the modulated waveform

(b) depth of modulation of modulated waveform

DSBC-AM ../9

- If V_x is peak value of signal

- The rms value is

$$V_{x_{rms}} = \frac{V_x}{\sqrt{2}}$$

- For a two or more tone modulating signal

$$V_{T_{rms}} = \sqrt{V_{1_{rms}}^2 + V_{2_{rms}}^2 + V_{3_{rms}}^2 + \ldots}$$

DSBC-AM ../10

Exercise 2x

- A carrier wave $v_c(t) = 15\sin(4x10^3)t\ \ Volts$ is amplitude modulated by the signal $v_m(t) = 3\sin\left(2x10^3\right)t\ + 2\sin(5x10^3)t\ \ Volts$.

  Determine

  (a) the rms value of the modulating signal

  (b) the depth of modulation of the modulated signal

DSBC-AM ../11

Example

a) A 1 MHz carrier with an amplitude of 1 V peak is modulated by a 1 kHz signal with $m = 0.5$. Sketch the amplitude spectrum.

b) An additional 2 kHz signal modulates the carrier with $m = 0.2$. Sketch the spectrum. What is the bandwidth in each case?

Solution

(a) The three frequency components will be

DSBC-AM ../12

(b) Additional modulating signal (f_{m2}) at different frequency

Power in DSBC-AM

- DSBC AM signal signal

$$v_{am}(t) = V_c \cos \omega_c t + \frac{mV_c}{2} \cos(\omega_c + \omega_c)t + \frac{mV_c}{2} \cos(\omega_c - \omega_c)t$$

- Power delivered to a resistance *R* is

$$P_{am} = P_c + P_{USB} + P_{LSB} = P_c + P_{SB}$$

$$= \left[\frac{V_c^2}{2R} + \frac{m^2 V_c^2}{4R} \right] = P_c \left(1 + \frac{m^2}{2} \right)$$

Power in DSBC-AM ../2

- Carrier Power

$$P_c = \frac{2}{2+m^2} P_{am}$$

- Total sideband power

$$P_{SB} = \frac{m^2}{2} P_c = \frac{m^2}{2+m^2} P_{am}$$

- for *m* = 1, P_{SB} = 50% of P_c

- as *m* reduces, P_{SB} reduces rapidly as a square law

Power in DSBC-AM ../4

- Transmission efficiency

$$\eta = \frac{P_{SB}}{P_{am}} = \frac{m^2}{2+m^2}100\%$$

- m = 1, η = 33.3%

Power in DSBC-AM ../4

Exercise 2x

- A DSBC system has a carrier frequency of 470 MHz and a modulating signal frequency of 5 kHz. Calculate the bandwidth of the transmitted signal. If *Vc* = 24 V, calculate the modulated carrier power for a load of 500 Ω when (a) *m* = 0, (b) *m* = 0.5, (c) *m* = 1